

DURHAM 11 W, NC, USA

WMO: 756940

Lat: 35.9705N

Lon: 79.0931W

Elev: 562

StdP: 14.40

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 03758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	22.0	-1.0	5.3	21.5	4.6	7.1	27.1	10.8	51.4	9.9	46.8	2.5	N/A	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	92.9	74.0	90.5	73.7	88.4	73.2	77.3	87.3	76.2	85.9	75.2	84.5	4.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	131.9	81.5	73.4	126.7	80.3	72.4	122.5	79.3	41.0	87.5	39.9	85.9	38.9	84.8	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.3	7.0	10.5	95.9	5.3	2.2	6.7	97.5	3.6	98.8	0.7	100.1	-3.1	101.7		
			9.7	79.0	4.5	1.4	6.5	79.9	3.8	80.7	1.3	81.5	-2.0	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.6	39.1	43.7	50.1	59.4	67.6	74.7	77.6	76.5	71.5	60.5	48.9	44.4
	DBStd	15.46	10.01	10.16	9.80	8.11	6.67	4.71	3.97	4.19	5.65	8.04	8.09	9.40
	HDD50	1085	357	229	124	16	0	0	0	0	0	12	115	232
	HDD65	3457	802	599	469	202	56	2	0	0	14	186	485	642
	CDD50	4582	20	52	127	299	545	742	856	821	645	336	82	57
	CDD65	1479	0	2	7	35	136	294	391	356	209	45	2	2
	CDH74	12459	0	10	62	354	1063	2645	3467	2955	1541	353	7	2
	CDH80	4657	0	0	3	61	286	1019	1469	1188	552	79	0	0
Wind	WSAvg	3.1	3.4	3.7	3.9	3.9	3.0	2.7	2.7	2.5	2.6	2.7	2.9	3.0
Precipitation	PrecAvg	46.7	4.0	3.7	4.4	3.3	4.1	4.1	4.3	4.6	3.9	3.5	3.4	3.3
	PrecMax	61.3	8.9	6.8	8.0	7.4	7.6	10.2	8.4	9.1	24.0	10.1	12.6	7.2
	PrecMin	33.3	1.2	0.6	1.5	0.7	1.1	0.9	0.3	0.8	0.1	0.3	0.6	0.3
	PrecStd	7.1	1.9	1.6	1.9	1.6	1.7	2.2	2.0	1.9	3.9	2.3	2.4	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.1	75.4	79.9	85.3	89.6	93.6	96.0	96.0	92.5	86.6	74.4	72.9
		MCWB	58.8	62.6	60.3	64.2	71.2	74.7	74.5	73.0	71.0	71.0	62.3	64.6
	2%	DB	64.6	70.3	75.3	81.8	86.0	90.9	92.9	92.0	88.9	81.8	70.7	68.5
		MCWB	55.7	58.3	59.2	63.8	69.8	73.9	74.7	73.9	70.3	68.5	60.8	62.4
	5%	DB	60.2	66.0	71.6	78.3	83.3	88.6	90.5	89.1	86.2	77.8	67.4	64.6
		MCWB	52.5	57.0	56.8	62.5	68.6	73.6	74.8	73.7	70.4	65.5	58.8	60.1
	10%	DB	55.6	61.6	66.8	74.7	80.3	86.1	87.8	86.7	83.3	74.4	63.6	60.8
		MCWB	48.7	53.7	55.2	61.1	67.7	72.5	74.1	72.9	69.7	64.6	55.7	55.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.0	64.2	64.4	70.6	74.1	77.4	79.3	78.4	75.5	74.3	66.8	67.0
		MCDB	67.1	71.3	73.4	77.7	83.1	88.5	89.7	87.6	83.3	79.6	70.3	70.1
	2%	WB	59.1	61.6	62.4	67.6	71.9	75.9	77.5	77.0	74.2	71.8	63.9	63.8
		MCDB	63.3	67.8	70.9	75.9	81.6	86.8	87.9	86.0	82.5	77.5	68.6	66.8
	5%	WB	53.5	58.8	60.4	65.5	70.8	74.8	76.5	75.6	72.8	69.0	61.2	60.5
		MCDB	57.0	65.0	67.9	73.3	80.3	85.4	86.8	83.8	80.9	75.7	65.9	64.1
	10%	WB	49.6	54.2	58.3	63.6	69.5	73.6	75.4	74.6	71.4	65.8	57.8	55.4
		MCDB	55.8	60.4	65.1	71.8	78.4	83.7	85.6	82.3	78.7	72.3	63.2	59.5
Mean Daily Temperature Range	5% DB	MDBR	19.0	20.3	21.3	22.5	20.0	20.2	19.2	18.7	18.8	21.2	21.4	18.4
		MCDBR	25.1	24.7	24.9	26.1	22.9	22.4	22.0	22.2	23.6	23.9	24.8	20.4
		MCWBR	17.7	15.9	12.9	11.4	9.7	8.7	7.6	7.4	8.7	11.9	15.3	15.0
	5% WB	MCDBR	20.3	22.2	21.6	19.8	19.2	20.5	19.4	17.6	18.4	18.9	18.6	17.7
	MCWBR	18.0	16.5	12.9	10.7	8.6	9.0	7.7	7.2	8.2	10.7	14.5	14.6	
Clear-Sky Solar Irradiance	taub	0.311	0.328	0.361	0.402	0.449	0.495	0.536	0.526	0.420	0.379	0.329	0.314	
	taud	2.504	2.447	2.394	2.280	2.182	2.095	1.997	2.010	2.304	2.396	2.497	2.528	
	Ebn at Noon	282	289	288	280	267	254	242	242	265	267	273	274	
	Edh at Noon	26	31	36	42	47	51	56	54	38	32	26	24	
All-Sky Solar Radiation	RadAvg	810	1025	1354	1702	1884	2022	1900	1725	1436	1166	889	698	
	RadStd	54	122	96	128	150	144	118	87	128	150	70	59	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+263	+178

Nomenclature: See separate page